Republic of the Philippines
Sorsogon State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
Bulan Campus
Zone 8, Bulan, Sorsogon
Tel. No.; 056 311-0103; Email Address: cict@sorsu.edu.ph

BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

Name: Jiselle P. Fuedan **Section:**

Course/Program: BSCS 1-1 **Date:** 09/15/26

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, testing, and building applications and algorithm-driven software systems	1. Software Engineer 2. Backend Developer
Systems & Cloud Infrastructure	Maintaining hardware provisioning, network virtualization, uptime, and deployment pipelines	1. Systems Administrator 2. DevOps Engineer
Data Science & AI Engineering	Extracting insights from structured/unstructured data and training predictive AI models	1. Data Scientist 2. Machine Learning Engineer
Cybersecurity & Threat Defense	Safeguarding network perimeters, conducting penetration testing, and managing incident response	1. Information Security Analyst 2. Penetration Tester
Information Systems & Enterprise IT	Aligning business workflows with technology solutions and enterprise software architectures	1. Business Systems Analyst 2. IT Manager

My Chosen Pathway (Part B)

I am most interested in: Software Engineering

Why I chose it:

I chose it because I want to create applications and websites in a way that allows me to express myself. I have already experienced creating a website when I was in senior high school, so I want to improve my knowledge and learn more about it. Eventually, I also want to create my own application.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Engineer	• Mastering Best Practices • Algorithm Basics • Code & Debunk • Learning Frameworks
Mid-Level	Mid-Level Developer	• Implementing Complex Features • Algorithm Efficiency • Unit Testing • Code Reviews
Senior-Level	Senior Engineer	• System Architecture Design • Scalability & Performance • Mentoring Juniors • Full Stack Responsibility
Expert	Technical Lead	• Guiding Technical Decisions • Team Leadership • Architectural Overviews • Cross-functional Collaboration
Leadership	Software Architect	• High-level System Design • Defining Standards • Integrating Complex Systems • Full Lifecycle Strategy

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Software Architecture	N	I plan to learn it through my college subject
Python	N	I plan to learn through my college subjects and online tutorial

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Data structures	N	I plan to learn it through my college subject
Deep learning Frameworks(PyTorch)	N	I plan to learn it through online tutorials and to my college subject
Statistical Modeling	N	I plan to learn it through my college subject

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	
Company	
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	
Required Skills Listed (3–5)	

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose Software Engineering because I experienced doing one of its jobs, which is web development. However, because we had limited time, I could not learn more about the other details involved in creating a website. Even so, I still enjoyed developing one, which is why I want to pursue and learn more about it.

2. Looking at your roadmap, which career stage excites you the most, and why?

In the roadmap, the one that excites me the most is the senior level, which is becoming a Senior Engineer. Even though there are higher levels, I think this is the most possible level I could attain. Additionally, my instructor mentioned that this stage has earnings that can reach six digits, which excites me even more.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

The skill or competency for this pathway that I already have, I think, is creativity. I am an imaginative person, and I have a wide imagination, which could help me when developing websites or applications. For example, when creating the theme of a website, my creativity could help me make it more interactive and interesting to users. The skill that would be hard for me to develop is fast learning because I tend to focus on the details before moving on to something else. If I don't understand a concept, I will spend time trying to understand it because I believe that it is better to fully understand something rather than just memorize it and move on. I think this can be challenging in Software Engineering because I may need to learn new programming languages, tools, and concepts quickly.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

The roadmap changed or confirmed my plan because it provides a clear progression of levels for the career. I also gained information about the skills and responsibilities I need to have to reach each level. This could help me set my goals and stay motivated.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date